AAKASH SHARMA

Senior Software Engineer | Python & Django | AI/LLM & Microservices

Gurugram, India | er.aakashtrikha@gmail.com | +91 98768 30251 | github.com/Aakash017 | linkedin.com/in/aakash-sharma-121ab5165

SUMMARY

Senior Software Engineer leading a 15-person engineering team at Wood Mackenzie, with 8+ years of experience building scalable Python/Django microservices that cut AWS infrastructure costs by 45% and system downtime by 70%. Hands-on with AI/LLM solutions including RAG pipelines and agent-based workflows (LangChain, LangGraph, A2A). Proven record of delivering backend systems, RESTful APIs, and Apache Airflow ETL pipelines across the energy, insurance, and automotive domains, with containerized deployments on AWS and GCP. Currently focused on productionizing agentic AI workflows for market intelligence.

WORK EXPERIENCE

Senior Software Engineer — **Wood Mackenzie** Gurugram, India | 11/2023 – Present

- Provide technical leadership across a cross-functional team of 15 (12 developers, 3 QA), driving architecture decisions, requirement gathering, and code quality standards.
- Built AI-powered solutions leveraging LLMs (OpenAI, Claude, AWS Bedrock, Gemini), RAG pipelines with vector databases, and agent-based workflows with LangChain, LangGraph, and A2A (agent-to-agent) protocols, automating previously manual analysis workflows.
- Boosted system scalability by 40% by designing and developing microservices in Python using Django and FastAPI with async I/O (asyncio) and Pydantic schema validation, enabling seamless integration across diverse market environments.
- Achieved a 99% accuracy rate in market data processing by developing a data quality checker using Python and the Django framework, backed by pytest unit and integration test suites.
- Reduced AWS infrastructure costs by 45% by profiling and optimizing Python code and right-sizing EC2, Lambda, and RDS usage.
- Centralized refinery data management by designing and developing an end-to-end Django web application, replacing fragmented manual workflows.

Software Engineer — **PasarPolis India Pvt Ltd** Gurugram, India | 11/2021 – 11/2023

- Defined and led the technical architecture strategy for Python/Django web-based solutions, establishing design standards adopted by a 6-engineer team.
- Resolved 100% of major production issues in 50 days against a 90-day mandate by leading team-wide triage and remediation; recognized with the Employee of the Month award.
- Reduced system downtime by 70% and improved overall performance by designing robust, scalable, and secure microservices in Django.
- Improved code quality and maintainability scores by 51%, measured via code-review metrics, by instituting rigorous Python review practices.
- Delivered ETL pipelines, database solutions, and end-to-end web and API services, orchestrating event-driven data workflows processing 23 million rows per day across 20+ pipelines with Apache Airflow, Celery, and RabbitMQ.

Software Engineer — **RenewBuy** Gurugram, India | 12/2019 – 06/2021

- Served as technical lead for a cross-functional team of 11 (8 developers, 3 QA), overseeing delivery of insurance platform features from design through release.
- Reduced manual data entry by 50%, saving 20+ hours per week, by leading the design and development of an operations panel in Python and Django.
- Increased fraud claim detection accuracy by 84% by spearheading the system design and development of a video inspection rule engine.
- Contributed to 60% business growth by engineering a policy renewal journey using Python and Django.

Python Developer — Bains Innovation Labs Chandigarh, India | 06/2018 – 12/2019

- Developed Django REST APIs for Autolife and Carsandjobs enabling third-party integrations, and increased completed transactions by 30% by integrating PayPal, Authorize.net, and Moneris payment gateways.

SKILLS

Languages & Frameworks: Python, Django, Flask, FastAPI, asyncio, Pydantic, RESTful APIs, SQL

AI & LLM: OpenAI, Anthropic Claude, AWS Bedrock, Gemini, RAG, Vector Databases (pgvector, FAISS), Embeddings, LangChain, LangGraph, A2A (Agent-to-Agent), AI Agents

Data & Messaging: Apache Airflow, Celery, RabbitMQ, MySQL, PostgreSQL, Redis, MongoDB, DynamoDB, ETL Pipelines

Cloud & DevOps: AWS (EC2, S3, Lambda, SQS/SNS, RDS, CloudWatch), GCP, Docker, Kubernetes, Terraform (IaC), CI/CD, Linux, Git, GitLab

Testing & Observability: pytest, unittest, Integration Testing, CloudWatch, Grafana, Prometheus

Security: OAuth2, JWT, REST API Security

Practices: Microservices, Event-Driven Architecture, System Design, API Design, Code Review, Agile/Scrum, Performance Optimization

EDUCATION

FCET Ferozepur — Bachelor of Engineering and Technology, Computer Science and Engineering
Ferozepur, India | 07/2011 – 08/2015

CERTIFICATIONS & AWARDS

- Apache Airflow using GCP
- Employee of the Month, PasarPolis (04/2022 – 05/2022)

Languages: English, Hindi